

Project Documentation: Password Strength Checker

1. Overview

The Password Strength Checker is a client-side web utility designed to provide immediate security feedback to users. By evaluating user input against industry-standard password complexity criteria, the application promotes the creation of robust credentials, enhancing overall user account security.

2. Security Logic & Scoring

The application utilizes a modular scoring system. A password is analyzed using Regular Expressions to identify the presence of specific character sets. The total score **S** determines the strength categorization:

Criteria	Weight
Minimum 8 Characters	+1
Contains Uppercase	+1
Contains Lowercase	+1
Contains Numeric	+1
Contains Special Character	+1

The calculated score **S** (ranging from 0 to 5) maps to the following UI feedback states:

Score	Strength Level	Color
1	Too Short	Red

2	Weak	Red
3	Fair	Orange
4	Good	Green
5	Strong	Green

3. Technical Implementation

Frontend Architecture:

- **Structure:** Semantic HTML5 elements for accessibility.
- **Styling:** CSS3 with transitions for smooth visual feedback and a responsive, card-based layout.
- **Interactivity:** Vanilla JavaScript (ES6+) for real-time DOM manipulation, password visibility toggling, and input event listening.

4. User Experience Features

- **Instant Feedback:** The strength bar updates synchronously as the user types, reducing cognitive load.
- **Visibility Toggle:** Users can toggle password visibility, reducing entry errors.
- **Visual Hierarchy:** High-contrast color indicators and clear labeling provide intuitive cues.

5. Future Enhancements

To further improve the utility for enterprise applications:

- **Dictionary Attack Prevention:** Integration with APIs to check against lists of known breached passwords.
- **Entropy Calculation:** Implementing actual Shannon entropy formulas for a more scientific measure of randomness.